A PERIODIC TWELVE-FOLD TILING BY THE SPECTRE

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Abstract. The Spectre of Smith, Myers, Kaplan and Goodman-Strauss tiles the plane by rotations and translations, and every such tiling is non-periodic. Andrew Bayly asked whether it can tile the plane periodically if each point is allowed to be covered n times, for some $n > 1$. It can: we give 54 rotated and translated copies of the Spectre whose translates by a lattice cover almost every point exactly 12 times. The proof is a finite check. The edges of the 54 copies cancel in oppositely oriented pairs modulo the lattice, so the covering number is constant, and comparing areas shows that the constant is 12. For the straight-edged Spectre the whole argument is formally verified in Lean.

1. The question

In 2023 Smith, Myers, Kaplan and Goodman-Strauss found a polygon, the Spectre, that tiles the plane using rotations and translations but only non-periodically, and they showed that replacing its straight edges by a suitable curve gives a strictly chiral tile with the same property [7]. A tiling covers almost every point once. In an *n-fold tiling* almost every point lies in exactly n tiles, overlaps being allowed. Andrew Bayly asked on MathOverflow [1] whether the Spectre admits a *periodic* n-fold tiling for some $n > 1$, that is, a multiple tiling invariant under a lattice of translations.

Theorem 1. *There is a periodic* 12*-fold tiling of the plane by copies of the Spectre under rotations and translations, with* 54 *copies in each period cell. This holds for the straight-edged Spectre and for every curved Spectre of the common-curve family.*

The tiling uses no reflections and no two of its tiles coincide. Adding generic translates of the whole configuration gives periodic tilings of every multiplicity $12m$. Whether some $1 < n < 12$ is possible is open.

Periodic multiple tilings have been studied mainly for one tile moved by translations alone: by convex or centrally symmetric polygons [5, 2], and, in the lattice $\mathbb{Z}^2$, by Greenfeld and Tao, who proved the periodic tiling conjecture for multi-tilings of $\mathbb{Z}^2$: a finite set that multi-tiles $\mathbb{Z}^2$ by translations also multi-tiles it periodically [3]. The Spectre needs twelve rotations and its vertices lie in a module of rank four, so neither line of work applies.

2. The construction

Let $\zeta = e^{\pi i/6}$. The straight-edged Spectre Tile$(1, 1)$ is the polygon with vertices $p_0 = 0$ and $p_{i+1} = p_i + \zeta^{d_i}$ for $0 \le i \le 13$, where

$$(d_0, \dots, d_{13}) = (0, 10, 1, 3, 0, 2, 5, 7, 4, 6, 6, 8, 11, 9).$$

Its fourteen edges have unit length and directions that are multiples of $30°$ (Figure 1). Every vertex lies in $\mathbb{Z}[\zeta]$, which is a free abelian group with basis $1, \zeta, \zeta^2, \zeta^3$, since $\zeta^4 = \zeta^2 - 1$.

A curved Spectre of the common-curve family [7, Lemma 2.1] replaces edge i by $p_i + \zeta^{d_i}\gamma(t)$ when i is even and by $p_{i+1} + \zeta^{d_i+6}\gamma(1-t)$ when i is odd, for a fixed simple curve γ from 0 to 1, chosen so that the boundary stays simple. Write S for the resulting tile. The regions added and removed along the edges cancel in pairs, so

$$|S| = |\text{Tile}(1, 1)| = 3 + 3\sqrt{3}.$$

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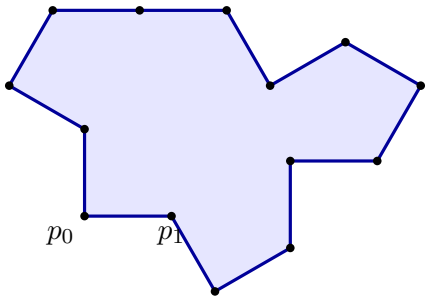


FIGURE 1. The straight-edged Spectre Tile$(1,1)$, traced counterclockwise from $p_0 = 0$. In a curved Spectre each edge is replaced by a copy of one curve, traversed forwards on even edges and backwards on odd ones.

Put

$$a = \zeta^2 + \zeta^3, \quad b = -2 - \zeta - \zeta^3, \quad v = 3 + \zeta + 2\zeta^2 + 3\zeta^3, \quad w = -4 - 3\zeta + 3\zeta^2 + \zeta^3,$$

and let $\Lambda = \mathbb{Z}v + \mathbb{Z}w$. The four vectors a, b, v, w form a basis of $\mathbb{Z}[\zeta]$ (their determinant in the basis $1, \zeta, \zeta^2, \zeta^3$ is 1). The tiles are

$$\zeta^r S + xa + yb + \lambda, \qquad \lambda \in \Lambda,$$

for the 54 triples (r, y, x) of Table 1.

r	y	values of x
0	5	$-2, -1, 2, 3$
1	-10	-8
2	-1	$-2, -1, \ldots, 5$
4	-10	$-5, -4, -3, -1, 0, 1$
4	-4	$-4, -3, 0, 1$
4	2	$-3, 1$
5	17	$-1, 3$
6	-13	$-9, -5$
6	-7	$-8, -7, -6, -4, -3, -2$
6	-1	$-3, 1$
7	8	$3, 4, 8$
8	-1	$-10, -9, \ldots, -3$
10	2	$-5, -1$
10	8	$-5, -4, -1, 0$

TABLE 1. The 54 placements in one period cell: the tile $\zeta^r S + xa + yb$ for each listed x.

3. Why the covering number is constant

For a point z off all tile boundaries let $N(z)$ be the number of tiles containing it. It is invariant under Λ, so it is a function on the torus $\mathbb{C}/\Lambda$.

Lemma 2. *The $54 \cdot 14 = 756$ edges of the tiles in one period cell split into 378 pairs. The two edges of a pair are the same rotated copy of the edge curve, at the same position modulo Λ, traversed in opposite directions.*

The lemma is a finite check, and it can be done in integers. Every vertex of every tile lies in $\mathbb{Z}[\zeta]$. The map $K\colon \mathbb{Z}^4 \to \mathbb{Z}^2$ given by

$$K = \begin{pmatrix} -2 & 3 & 0 & 1 \\ 1 & 2 & 5 & -5 \end{pmatrix}$$

in the basis $1, \zeta, \zeta^2, \zeta^3$ satisfies $Ka = (1, 0)$, $Kb = (0, 1)$ and $\ker K = \Lambda$, so K reads off the position of a point of $\mathbb{Z}[\zeta]$ modulo Λ. Record edge i of the tile (r, x, y) as

$$\Big((r + d_i + 6\varepsilon_i) \bmod 12, \quad (x, y) + K\big(\zeta^r p_{i+\varepsilon_i}\big)\Big), \qquad \varepsilon_i = i \bmod 2,$$

with sign $(-1)^i$. The first entry says which of the twelve rotated copies of the curve the edge is, and the second where that copy starts modulo Λ; the sign says in which direction it is traversed. The lemma says that the 378 positive records and the 378 negative records are the same multiset, and a direct count confirms it.

Proof that N is constant. Off the tile boundaries, which form a null set, N is locally constant. Crossing an edge of a tile from its outside to its inside raises that tile's contribution by one. By Lemma 2 every edge of the period cell is matched, modulo Λ, by the same curve traversed the other way, which belongs to a tile entered from the opposite side, so the two changes cancel and N takes the same value on the two sides of every edge. The torus is connected, so N is constant almost everywhere. □

For the straight Spectre this argument is made precise in Lean through a triangulation: each tile is cut into twelve triangles, the indicator of a triangle is a sum of signed "shadows" of its three edges, and the 1944 triangle edges cancel modulo Λ (Section 5). For a curved Spectre the step from the cancelling records to the jump of N across each edge uses the Jordan curve theorem for the edge curve.

4. Why the constant is twelve

In Cartesian coordinates,

$$v = \Big(4 + \tfrac{\sqrt{3}}{2},\ \tfrac{7}{2} + \sqrt{3}\Big), \qquad w = \Big(-\tfrac{5}{2} - \tfrac{3\sqrt{3}}{2},\ -\tfrac{1}{2} + \tfrac{3\sqrt{3}}{2}\Big),$$

so a period cell has area $\det(v, w) = \frac{27}{2}(1 + \sqrt{3})$. Integrating the constant N over a cell counts each of the 54 tiles once by area:

$$N = \frac{54\,|S|}{\det(v, w)} = \frac{54 \cdot 3(1 + \sqrt{3})}{\frac{27}{2}(1 + \sqrt{3})} = 12.$$

This proves Theorem 1. Figure 2 counts the tiles over a fine grid of points in four period cells for one curved Spectre.

Remark. The construction was found by a computer search for a set of placements whose edge records cancel. Such a cancellation forces a constant covering number on its own, which is why no matching rules or local configurations of Spectre tilings enter the argument.

5. Formal verification

For the straight-edged Spectre, Theorem 1 is proved in Lean 4 with Mathlib [6]: for the 54 placements and every vector of Λ, almost every point of the plane lies in exactly 12 tiles. The proof triangulates the tile, shows in general that a periodic family of triangles whose edges cancel modulo the lattice has an almost everywhere constant count, and computes the constant from the areas. The finite facts (the cancellation of the 1944 triangle edges, the orientation of every triangle, the triangulation's boundary, the distinctness of the placements) are checked by the kernel, and the curved-edge records of Lemma 2 are checked the same way. The step from those records to constant coverage for a curved Spectre, which needs the Jordan curve theorem, is not formalised. Everything uses only Lean's standard axioms, and planted errors in the statements are rejected. A program checks the records and the areas in exact integer arithmetic, an independent one counts the tiles at random points, and the code and the Lean files are at [4].

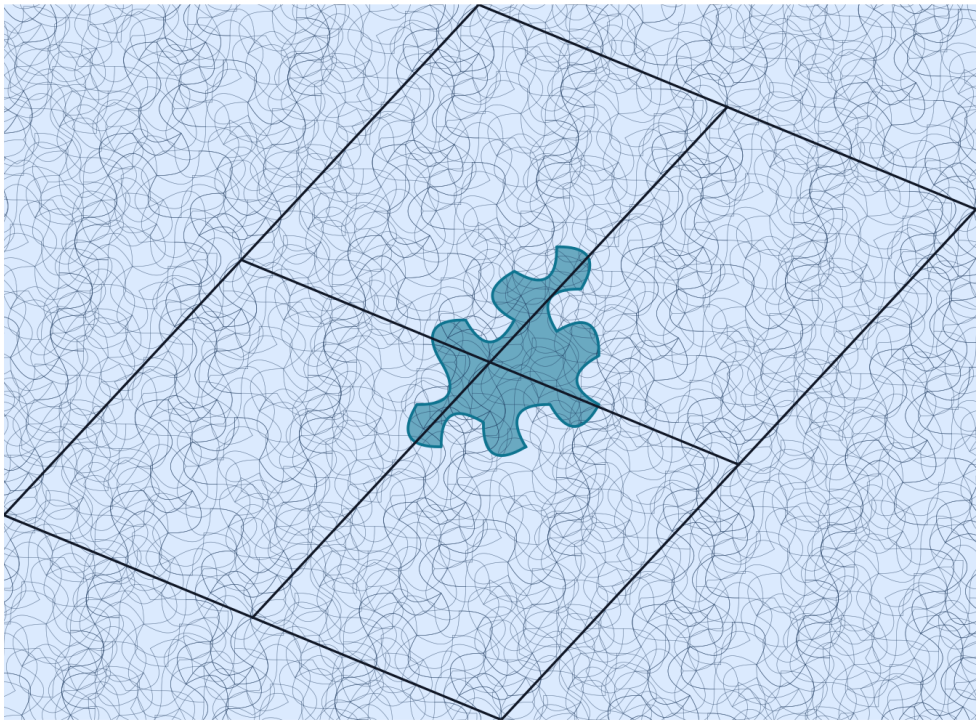


FIGURE 2. Four period cells (black lines) of the cover by a curved Spectre, with one tile shaded. Every one of the 359,800 sample points of the grid lies in exactly 12 tiles.

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